DESIGN AND DEVELOPING THE SECURE INTRUSION DETECTION SYSTEM DOOR LOCK SYSTEM USING RFID OVER NETWORKS

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA0712 & Computer Networks

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BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

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(COMPUTER SCIENCE OF ENGINEERING , AI&ML)

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "DESIGN AND DEVELOPING THE SECURE INTRUSION DETECTION SYSTEM DOOR LOCK SYSTEM USING RFID OVER NETWORKS" has been carried out by C.Sravanthi (192525463), P.Krishna Priya(192511279) and A.G.Keshika Lakshmi(192521056) under the supervision of Dr.S.Salini and is submitted in partial fulfilment of the requirements for the current semester of the B.TECH AI&ML program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, [C. Sravanthi,P. Krishna Priya and A.G. keshika Lakshmi]of the [CSE,AI&ML AND IT], SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "DESIGN AND DEVELOPING THE SECURE INTRUSION DETECTION SYSTEM DOOR LOCK SYSTEM USING RFID OVER NETWORKS" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing& Analysis Contribution	Report Contribution	Approx. Contribution (%)
C.Sravanthi /192525463	RFID hardware integration,RFID reader/tag configuration, microcontroller interfacing, electronic door-lock integration	Designed and developed the RFID-based authentication and door-lock control section; integrated RFID reader, controller and lock mechanism	Tested authorized/unauthorized RFID access, lock response and authentication operation	Abstract, Introduction, Design & Development, Hardware Implementation sections	**34%**
P.Krishna Priya/192511279	Network communication, intrusion detection logic, unauthorized-access monitoring and alert mechanism	Developed the network-based communication and intrusion-detection functionality; implemented event reporting/monitoring concept	Tested unauthorized access attempts, repeated invalid attempts, network communication and intrusion-event detection	Literature Review, Network/Intrusion Detection, Testing & Analysis sections	**33%**
A.G.Keshika Lakshmi /192521056	System monitoring, result analysis, performance evaluation, documentation and project coordination	Contributed to overall system integration, monitoring workflow and improvement of the proposed security architecture	Analysed system behaviour, authentication results, security response and overall performance	Results, Conclusion, Future Work, References and final report compilation	**33%**
Total					**100%**

ABSTRACT

Unauthorized physical access is a significant concern in homes, laboratories, offices and restricted facilities. Conventional mechanical locks provide limited capability for electronic identification, access logging and remote monitoring. This project presents a secure RFID-based door-lock system that integrates user identification, electronic locking, network communication and intrusion detection. An RFID reader captures the identifier of a presented tag, while a microcontroller validates the identifier against the authorized list. If the tag is valid, the controller drives the electronic locking mechanism and records the access event. If the tag is invalid, the door remains locked and the failed attempt is recorded. Repeated invalid attempts can be treated as intrusion events and reported through the network monitoring layer. The system is designed with local access-control logic so that the basic locking decision does not depend completely on network availability, while the network is used for status reporting, event logging and alert generation. Implementation involves RFID interfacing, embedded programming, actuator control, network connectivity and system integration. Testing covers authorized access, unauthorized access, repeated invalid attempts, lock response, network event reporting and recovery after interruption. The proposed prototype provides a low-cost and modular foundation for networked physical access control. Its principal contribution is the integration of RFID authentication and intrusion monitoring with a network-connected electronic door lock, while also considering practical constraints such as cost, power, reliability, safety and cybersecurity.

Keywords: RFID, Door Lock, Intrusion Detection, Network Security, Access Control, IoT

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
RFID	Radio Frequency Identification	—
IDS	Intrusion Detection System	—
IoT	Internet of Things	—
MCU	Microcontroller Unit	—
UID	Unique Identifier	—
LAN	Local Area Network	—
IP	Internet Protocol	—
GPIO	General Purpose Input/Output	—
LED	Light Emitting Diode	—
API	Application Programming Interface	—
MAC	Media Access Control	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

RFID provides contactless identification by allowing a reader to obtain information from an RFID tag. In an electronic door-lock application, the RFID reader acts as the first security interface. A controller receives the tag identifier, checks whether it is authorized and then controls an electronic lock. Adding a network interface extends the system beyond a standalone lock by allowing access events, denied attempts and intrusion conditions to be monitored remotely. The proposed project therefore combines embedded hardware, RFID authentication, electronic actuation and network security into one integrated system.

1.2 Need for the Project

Traditional keys can be lost, duplicated or shared and normally provide no automatic record of who attempted to enter. A standalone RFID lock improves convenience but may still lack centralized monitoring and detection of repeated unauthorized attempts. A network-enabled design can provide event visibility and faster awareness of suspicious activity. The engineering need is to balance security, response time, cost, power consumption, reliability and ease of implementation.

1.3 Problem Statement

Design and develop a secure intrusion-detection door-lock system using RFID over a network that authenticates users, controls an electronic locking mechanism, detects unauthorized access attempts and communicates security events to a monitoring layer while operating reliably under practical hardware, software, power and network constraints.

1.4 Project Aim

To design, develop and validate a network-connected RFID door-lock system that provides controlled access and detects unauthorized entry attempts.

1.5 Project Objectives

1. Design the complete RFID-controller-lock architecture.
2. Develop reliable RFID authentication logic.
3. Interface the controller with the electronic door-lock actuator.
4. Implement network communication for access-event monitoring.
5. Detect repeated unauthorized attempts and generate intrusion events.
6. Test and evaluate system functionality, reliability and security.

1.6 Scope

The project includes RFID tag identification, authorization checking, controller-based lock control, network connectivity, access-event logging, intrusion-event detection and prototype-level verification. The work does not claim commercial security certification, resistance to every RFID-cloning technique or protection against all possible network attacks. The prototype assumes a regulated low-voltage power supply and a configured local network.

1.7 Expected Engineering Outcome

The expected outcome is a working prototype in which an authorized RFID tag unlocks the door for a controlled period, an unauthorized tag is rejected, repeated invalid attempts can generate an intrusion alert, and access events are communicated to a monitoring system over the network.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review considers RFID identification, electronic access-control systems, embedded door locks, IoT/network security and intrusion detection. Existing approaches are compared according to authentication capability, monitoring, cost, reliability and security limitations.

2.2 Review of Existing Work

RFID-based access-control systems commonly use a reader, a tag and a controller. The controller compares the received UID or credential with an authorization list and operates an actuator when access is permitted. Networked IoT versions add communication and monitoring functions. Key engineering concerns include protecting credentials, preventing unauthorized access, maintaining operation during network failure and ensuring that the actuator behaves safely. Intrusion detection can improve security by counting invalid attempts and reporting suspicious activity rather than simply denying access.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
Mechanical lock	Simple and inexpensive	No electronic identity or event logging	Baseline
Standalone RFID lock	Convenient and fast	Limited remote monitoring	Authentication layer
Keypad/PIN lock	No physical RFID tag needed	PIN sharing/observation possible	Alternative
Networked RFID with IDS	Authentication, monitoring and al	Higher complexity	Proposed solution

2.4 Research / Engineering Gap

Low-cost access-control prototypes often focus on the unlock function and provide limited treatment of network monitoring, repeated failed attempts, security-event logging and recovery. The engineering gap addressed here is the integration of physical access control with network-aware intrusion detection while retaining a simple prototype architecture.

2.5 Proposed Contribution

The proposed system links RFID authentication directly to door actuation and adds network reporting of access and intrusion events. The primary authorization decision remains local to the controller, reducing dependence on continuous network availability, while the network provides monitoring and evidence of security events.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Requirement	Type	Measurable Target
Recognize authorized RFID tag	Functional	Authorized tag accepted
Reject unauthorized tag	Functional	Door remains locked
Control electronic actuator	Functional	Lock changes state after authorizati
Transmit access event	Functional	Event reaches monitoring layer
Detect repeated invalid attempts	Functional	Intrusion event generated
Safe low-voltage operation	Non-Functional	No exposed hazardous

on

3.2 Constraints & Applicable Standards

Constraints include component cost, controller memory and processing capacity, actuator current requirements, RFID read range, network availability, power stability and physical enclosure limitations. Relevant technology standards include ISO/IEC 14443 for proximity/contactless cards where applicable to the selected RFID technology and IEEE 802.3/802.11 concepts where Ethernet/Wi-Fi networking is used. Final compliance should be checked against the exact modules and deployment environment.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
RFID identification	Reliable detection of authorized tag	UID/event	High
Controller	Sufficient GPIO and communication interfaces	—	High
Lock actuator	Controlled electronic locking/unlocking	state	High
Network	Local IP communication	—	High
Alert mechanism	Log/report unauthorized attempts	event	High
Power	Regulated supply for all modules	V/A	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Standalone RFID lock. This approach is simple and low cost but provides little remote visibility or intrusion reporting.

Alternative 2: Networked RFID door lock with intrusion detection. This approach adds event monitoring and alerts while keeping the physical access decision at the controller.

Alternative 3: Multi-factor RFID plus PIN/biometric system. This can improve authentication strength but increases cost and implementation complexity for the prototype.

Criterion	Standalone RFID	Networked RFID + IDS	Multi-factor
Performance	High	High	High
Cost	High advantage	Medium	Low
Monitoring	Low	High	High
Security	Medium	High	Very high
Implementation complexity	Low	Medium	High

3.5 Selected Design & Detailed Design

The networked RFID + IDS architecture is selected because it provides the best balance between security functionality, cost and prototype feasibility. The RFID reader sends the detected tag information to the microcontroller. The controller performs authorization, drives the electronic lock and creates an event. The network interface forwards the event to a monitoring or alert layer. If an unauthorized tag is detected repeatedly, the intrusion logic raises an alert according to the configured threshold.

System Architecture: RFID Tag □ RFID Reader □ Microcontroller □ Authentication Decision □ Electronic Lock; Microcontroller □ Network Interface □ Monitoring / Alert Log.

3.6 Trade-off Analysis

Keeping authentication local improves availability when the network is unavailable, while network monitoring provides security visibility. More complex authentication could reduce the risk of credential misuse but would increase cost and integration effort. The selected design therefore prioritizes a reliable local decision, network event reporting and extensibility.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Evidence Evaluated	Impact on Decision
Sustainability	Low-power controller and reusable prototype components	Reduced energy and material waste
Ethics	Authorized-access policy and responsible event logging	Limit collected user information
Safety	Low-voltage wiring and protected actuator	Safer prototype operation
Security	Credential validation, failed-attempt detection, network protection	Improved misuse detection

3.8 Risk Register

Risk	Likelihood	Severity	Mitigation	Residual Risk
RFID credential misuse	Medium	High	Authorization control and monitoring	Medium
Network failure	Medium	Medium	Local access decision	Low/Medium
Power interruption	Medium	High	Regulated supply and safe lock state	Medium
Repeated unauthorized attempts	Medium	High	Attempt counter and alert	Low/Medium
Actuator failure	Low/Medium	High	Functional testing and recovery	Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The prototype was developed incrementally. First, the RFID reader and controller interface were established and verified. Next, the authentication logic and electronic lock control were implemented. Network communication was then added for access-event monitoring, followed by intrusion-event handling. Finally, the integrated system was tested under authorized, unauthorized and failure conditions.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Component / Tool	Purpose	Stage Used
RFID reader and tags	User identification	Authentication
Microcontroller board	Authentication and control logic	Core implementation
Electronic lock / servo / solenoid	Physical door control	Actuation
Relay/driver circuit	Interface actuator safely	Hardware integration
Network interface	Event/status communication	Monitoring
Embedded IDE / firmware tools	Program and debug controller	Development
Serial/network monitoring tool	Verify messages and events	Testing

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Authorized access	Present registered RFID tag multiple times	Door unlocks and event is logged
Unauthorized access	Present unregistered tag	Door remains locked
Repeated invalid access	Present invalid tags repeatedly	Intrusion event/alert is generated
Actuator operation	Trigger valid authorization	Lock reaches intended state
Network reporting	Generate access/denial event	Monitoring layer receives event
Recovery	Interrupt/restart network or controller	System returns to safe state

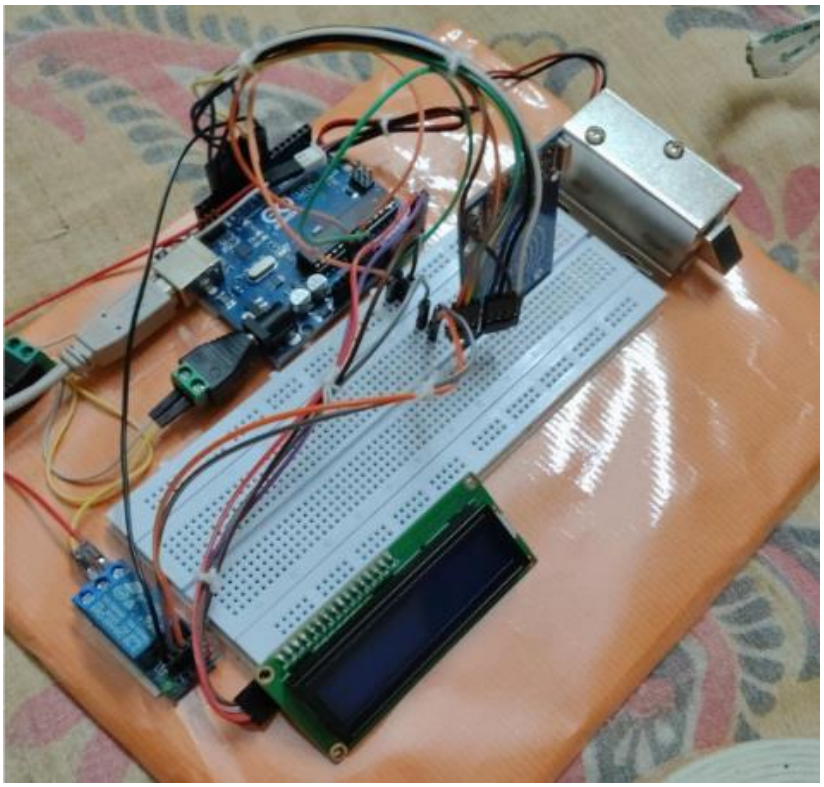
4.4 Results

The following result table is structured for the actual prototype observations. The report does not invent numerical measurements that were not supplied. During final testing, record the number of successful RFID reads, unauthorized denials, lock-actuation successes, network-delivery successes, response times and recovery outcomes.

Test	Expected Result	Observed Result	Status
Authorized RFID	Unlock + log	Enter measured result	Pass/Fail
Unauthorized RFID	Remain locked + log	Enter measured result	Pass/Fail
Repeated invalid attempts	Intrusion alert	Enter measured result	Pass/Fail
Network event	Event received	Enter measured result	Pass/Fail
Recovery	Safe return to operation	Enter measured result	Pass/Fail

4.5 Data Analysis & Engineering Judgment

The most important performance measures are authentication success rate, false acceptance/rejection observations, lock response time, event-delivery success and recovery behavior. Any failed test should be traced to RFID reading, wiring/power, firmware, actuator or network causes. This classification helps distinguish a component-level fault from a system-level design limitation.



4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design RFID access architecture	Reader-controller-lock integration	Fully / verify
Develop authentication logic	Authorization and denial logic	Fully / verify
Implement network monitoring	Access-event reporting	Fully / verify
Detect intrusion events	Failed-attempt detection	Fully / verify
Validate reliability	Defined test plan and results table	Insert actual results

4.7 Strengths & Limitations

Strengths: RFID-based identification; local access decision; electronic lock control; network event monitoring; repeated-failure detection; modular architecture; low-cost prototype approach.

Limitations: RFID credential security depends on the selected RFID technology; network communication requires correct security configuration; prototype mechanics may not represent commercial door hardware; actual performance depends on the measured environment and components.

4.8 Project Planning, Roles & Budget

Milestones: requirements □ architecture □ hardware assembly □ RFID firmware □ lock integration □ network communication □ intrusion logic □ system integration □ testing □ analysis □ documentation.

Budget Item	Estimated Cost (INR)
RFID reader + tags	500
Microcontroller/network board	800
Electronic lock/actuator + driver	700
Power supply and wiring	500
Prototype board/enclosure/miscellaneous	700
Approximate total	3200

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project presents a network-connected RFID door-lock system designed to improve electronic access control through identification, controlled actuation and intrusion-event monitoring. The proposed system authenticates RFID tags locally, operates an electronic lock for authorized users, rejects unauthorized users and communicates security events through a network monitoring layer. The design addresses practical engineering considerations including cost, power, reliability, safety and cybersecurity. Validation should be based on the defined test plan and the measured prototype results. Overall, the project demonstrates how embedded RFID technology and network communication can be integrated to create a practical foundation for secure physical access control.

5.2 Future Work

Future versions can incorporate multi-factor authentication, stronger cryptographic RFID credentials, encrypted network communication, tamper sensors, battery backup, secure event storage, mobile notifications, role-based access management and improved mechanical enclosure design. Longer-duration testing and controlled security testing against credential replay, cloning and network threats can further strengthen the design.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired: RFID interfacing, embedded authentication, actuator control, network communication, event logging and security testing.
Engineering & Professional Skills Developed: hardware integration, firmware debugging, test planning, data interpretation, documentation and teamwork.

Student 1 Reflection: The major learning was the integration of RFID hardware, controller logic and the electronic lock. The student developed practical skills in wiring, authentication and hardware testing. A key engineering decision was to keep the access decision local so that network interruption would not automatically prevent safe local operation. If repeated, stronger enclosure and tamper detection would be added.

Student 2 Reflection: The major challenge was implementing network communication and intrusion-event handling. The student gained experience in communication debugging, event logging and unauthorized-access testing. Separating access control from network monitoring improved system resilience. A future version would strengthen encrypted communication and credential protection.

Student 3 Reflection: The main learning was system integration, verification and technical documentation. The student developed skills in defining acceptance criteria, analyzing test outcomes and identifying limitations. The project showed the importance of testing both normal and abnormal operating conditions. A future version would add automated measurement and long-duration reliability testing.

REFERENCES

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[3] IEEE, *IEEE Std 802.3, Ethernet Standards*.

[4] IEEE, *IEEE Std 802.11, Wireless LAN Medium Access Control and Physical Layer Specifications*.

[5] ISO/IEC 14443, *Identification cards — Contactless integrated circuit cards — Proximity cards*.

[6] ISO/IEC 18000 series, *Information technology — Radio frequency identification for item management*.

[7] OWASP Foundation, *OWASP Internet of Things security guidance*.

[8] National Institute of Standards and Technology (NIST), IoT device cybersecurity guidance and recommendations.

[9] Arduino, *Arduino Documentation*, embedded hardware and software reference resources.

[10] Espressif Systems, *ESP32 Series Documentation*, where an ESP32-based network controller is used.

APPENDICES

Appendix A – Detailed Calculations
Include power-budget, actuator-current, communication-timing and any statistical/reliability calculations used.

Appendix B – Engineering Drawings / Schematics
Insert the final circuit schematic, wiring diagram, architecture diagram and enclosure drawing.

Appendix C – Source Code / Code Repository Information
Include essential firmware excerpts; maintain complete source code separately in the approved repository.

Appendix D – Datasheets
Attach datasheets for the RFID reader, controller, actuator/driver, power supply and network module.

Appendix E – Experimental / Raw Data
Insert the actual RFID read counts, response times, access decisions, network event logs and recovery observations.

Appendix F – Ethics / Institutional Approval
Include where applicable.

Appendix G – Project Meeting / Progress Records
Include meeting dates, decisions and action items.

Appendix H – User/Industry Feedback
Include where applicable.

Appendix I – Publications / Patents / Competitions / Product Outcomes
Include where applicable.

Appendix J – Individual Contribution Evidence
Include photographs, code records, test sheets or other evidence supporting each student’s contribution.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)